



ASSIGNMENT 4 – NM ; MARKS: 20

DUE DATE: **20th January 2025**

Question 1.

In machine learning, certain algorithms, like gradient descent, can be modeled using differential equations to track the change in error over time. Consider the model:

$$y' = \frac{1+t}{1+y} ; y(1) = 2 \quad \text{in} \quad 1 \leq t \leq 2$$

where t is time (in iterations), and $y(t)$ represents the error (loss) during training. The actual solution is:

$$y(t) = \sqrt{t^2 + 2t + 6} - 1$$

- a. Approximate $y(2)$ using:
 - a. **Euler's Method** ($n = 5$)
 - b. **RK-2 Method** ($n = 5$)
 - c. **RK-4 Method** ($n = 2$)
- b. Compute the values of **actual error** at $t = 2$ by comparing with the actual solution.
- c. Compare the methods and explain which method is more accurate and why?

Question 2.

Water flows from an inverted conical tank with circular orifice (opening) at the rate

$$\frac{dx}{dt} = -0.6\pi r^2 \sqrt{2g} \frac{\sqrt{x}}{A}$$

where r is the radius of the orifice, x is the height of the liquid level from the vertex of the cone, and A is the function of x and it represents the area of the cross section of the tank x units above the orifice. Suppose $r = 0.1 \text{ ft}$, $g = 32.1 \text{ ft/s}^2$, and the tank has an initial water level of 8 ft . The initial volume given by $V = \frac{1}{3}(xA)$ is $512(\frac{\pi}{3}) \text{ ft}^3$.

- a. Use the RK-4 Method to find the water level after 1 min with $h = 20 \text{ s}$.
- b. Will the tank be empty at any time during the first 1 minute ?